Basics & Reasoning Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Segment addition

ORIGINAL PROBLEM

B is between A and C. AB = 3x + 2, BC = 2x − 1, and AC = 36. Find AB.

INCORRECT WORKED SOLUTION

$3x + 2 = 2x - 1$

$x = -3$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Midpoint

ORIGINAL PROBLEM

M is the midpoint of AB, and AB = 26. If AM = 4x − 3, find x.

INCORRECT WORKED SOLUTION

$AM = AB$

$4x - 3 = 26$

$x = \frac{29}{4}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Complementary angles

ORIGINAL PROBLEM

∠A and ∠B are complementary. m∠A = (3x + 10)° and m∠B = (2x)°. Find x.

INCORRECT WORKED SOLUTION

3x + 10 + 2x = 180

5x + 10 = 180

5x = 170

x = 34

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Linear pair

ORIGINAL PROBLEM

Two angles form a linear pair and measure (5x + 10)° and (3x + 10)°. Find x.

INCORRECT WORKED SOLUTION

5x + 10 = 3x + 10

2x = 0

x = 0

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Converse

ORIGINAL PROBLEM

Write the converse of: "If a figure is a square, then it is a rectangle."

INCORRECT WORKED SOLUTION

Converse: If a figure is not a square, then it is not a rectangle.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Contrapositive

ORIGINAL PROBLEM

Write the contrapositive of: "If it rains, then the game is canceled."

INCORRECT WORKED SOLUTION

Contrapositive: If it does not rain, then the game is not canceled.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Counterexample

ORIGINAL PROBLEM

Is the statement "All prime numbers are odd" true or false?

INCORRECT WORKED SOLUTION

3 is prime and odd.

5 is prime and odd.

7 is prime and odd.

The statement is true.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Inductive vs. deductive

ORIGINAL PROBLEM

Test: 1 + 3 = 4, 3 + 5 = 8, 5 + 7 = 12 (all even). Is "the sum of two odd numbers is even" proven?

INCORRECT WORKED SOLUTION

Three examples all gave even sums.

So it is proven for all odd numbers.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Law of detachment

ORIGINAL PROBLEM

Given: If a number is divisible by 6, then it is divisible by 3. The number 15 is divisible by 3. What can you conclude?

INCORRECT WORKED SOLUTION

If divisible by 6, then divisible by 3.

15 is divisible by 3.

So 15 is divisible by 6.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Biconditional

ORIGINAL PROBLEM

Write as a biconditional, if possible: "If $x = 5$, then $x^2 = 25$."

INCORRECT WORKED SOLUTION

$x = 5$ if and only if $x^2 = 25$.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)